DFT HANDS-ON DAY

Connecting VASP, Machine Learning, and Experimental Data

Graduate Materials Simulation

Run / inspect DFT → **Quantify model error** → **Learn a correction with ML** → **Compare with experiment**

Today is not about proving that DFT is "good" or "bad."

It is about learning how simulation accuracy depends on the material and property, how experimental data exposes those limitations, and how machine learning can sometimes correct systematic model discrepancy.

Hands-On Day Objectives

By the end of today, you should be able to:

- Run or interpret a small VASP calculation and identify the simulation output needed for a materials-property prediction.
- Compare a DFT prediction directly with an experimental reference and quantify the error rather than judging agreement qualitatively.
- Explain why the accuracy of a DFT approximation such as PBE depends on the material and the property being predicted.
- Build a simple supervised machine-learning correction model using delta learning: the model learns the difference between DFT and experiment rather than replacing the DFT calculation.
- Use a training/test separation to determine whether the learned correction generalizes to materials that were not used to fit the model.
- Decide whether adding ML actually improves a physics-based prediction, including recognizing cases where ML provides little benefit or makes a prediction worse.

Core idea for the day

Simulation, machine learning, and experiment play different roles. DFT provides a physics-based prediction. Experiment provides the external benchmark. ML is used as a correction layer only when the data show that such a correction is useful.

How This Fits the Course

The course is built around connecting simulation methods to machine learning / AI and experimental characterization. Today combines all three pieces in a single workflow:

Course component	What you practice today
Simulation methods	Use VASP/PBE as a first-principles model and interpret the physical meaning of its outputs.
Experimental connection	Treat measured properties as a separate data source with its own meaning, uncertainty, and role as an external benchmark.
ML / AI	Use supervised regression to learn systematic model discrepancy rather than treating ML as a black-box replacement for physics.
Scientific judgment	Compare baselines, evaluate held-out performance, and decide whether the more complicated workflow is actually better.

CHAPTER 1

Correcting DFT Electronic Gaps with Machine Learning

Scientific question

Is PBE equally accurate for every material, and can a machine-learning model learn when a DFT electronic-gap prediction needs a large correction, a small correction, or essentially no correction?

Why Two Live Materials?

You will run the same general type of PBE electronic-structure calculation for two very different cases:

Silicon (Si)	Aluminum (Al)
A semiconductor. PBE captures the existence and qualitative nature of the electronic gap, but its numerical gap is substantially smaller than the room-temperature experimental value.	A metal. PBE correctly predicts gapless metallic electronic behavior. This is an example where the essential experimental classification is already captured without an ML correction.

The contrast is intentional. The lesson is not “DFT is bad.” The lesson is that the size and type of DFT error depend on the material and the property.

Workflow

Run Si + Al with VASP → **Extract electronic gap / metallicity** → **Compare PBE with experiment** → **Train delta-ML on other materials** → **Apply to held-out Si + Al** → **Ask whether ML helped**

Files You Will Use

- live_vasp/Si/ and live_vasp/Al/: VASP input folders containing POSCAR, INCAR, KPOINTS, POTCAR, and submit.sh; VASP generates EIGENVAL and other outputs.
- data/*.csv: training materials plus descriptors and experimental references for the two held-out live materials. Si and Al are excluded from training.
- Python files: vasp_gap_helper.py is a supplied parser; bandgap_ml_student_template.py is the guided analysis script you complete.

CHAPTER 1 - WORKFLOW

Step-by-Step: Electronic Gaps

1. Submit the two VASP calculations

From the Si directory:

```
qsub submit.sh
qstat -u $USER
```

Repeat for Al. These are intentionally small calculations, so the goal is to spend class time interpreting the outputs rather than waiting for compute.

2. Extract the PBE electronic behavior

The supplied helper reads EIGENVAL and identifies whether the sampled electronic structure is metallic; if it is not metallic, it reports the highest occupied state, lowest unoccupied state, and their energy difference.

Checkpoint

Before doing any ML, write down what PBE says about each material and compare it directly with the provided experimental reference. Which case is already described well? Which case has a large quantitative error?

3. Inspect the training data before fitting a model

Plot the PBE gap on the x-axis and the experimental gap on the y-axis. Add the ideal y = x line. Do not train a model yet.

- Are all materials equally far from the ideal line?
- Do metallic and semiconducting materials show the same kind of discrepancy?
- Would adding one constant value to every PBE prediction make sense?

4. Define the ML target as a correction

Instead of asking ML to predict the experimental property from scratch, use delta learning:

$$\Delta E_{gap} = E_{gap}(exp) - E_{gap}(PBE)$$

The model therefore learns how much the physics-based prediction should be corrected for a material with a given set of descriptors.

5. Train and evaluate the correction model

Use the provided train/test workflow in the template. Your first comparison is not between two ML algorithms. It is between:

Baseline	ML correction
Do nothing. Use the raw PBE prediction. For the correction target, this is equivalent to predicting $\Delta E_{gap} = 0$.	Fit a simple regression model to ΔE_{gap} using the PBE result and material descriptors.

Evaluate mean absolute error (MAE) on data that were not used to fit the model.

6. Apply the model to live Si and Al

Si and Al are held out from the training set. Use your live VASP result as an ML feature, predict the correction, and compute:

$$E_{gap}(\text{ML-corrected}) = E_{gap}(\text{PBE}) + \Delta E_{gap}(\text{ML})$$

7. Decide whether ML helped

For each live material calculate the absolute error before and after ML. A scientifically useful correction should improve the held-out prediction; it is entirely possible for ML to provide little benefit or even make an already-good result worse.

CHAPTER 1 - INTERPRETATION

What You Should Learn from Problem 1

Concept	Takeaway
DFT/PBE	A converged first-principles calculation can be physically meaningful while still having a material-dependent quantitative error.
Electronic structure	“Correct” can mean different things: getting metallic vs semiconducting behavior right is different from reproducing a semiconductor gap quantitatively.
Machine learning	A delta model can learn model discrepancy rather than replacing the underlying physics calculation.
Validation	Performance on the training set is not enough. Si and Al are useful because they test the model on materials withheld from fitting.
Experiment	Experimental data remain the external reference used to decide whether either the DFT or ML-corrected prediction is actually better.

Minimum Outputs

- A PBE-vs-experiment scatter plot for the training materials with the y = x reference line.
- A reported baseline error and ML-corrected error on held-out data.
- A final Si/Al comparison showing experiment, raw PBE, and ML-corrected predictions.
- A short conclusion answering: Did ML help Si? Did it help Al? Why should these answers not have to be the same?

CHAPTER 2

Predicting Experimental Lattice Constants from DFT

Scientific question

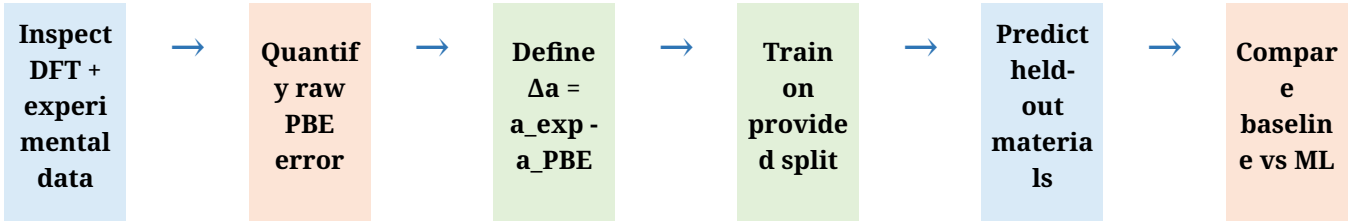
If PBE lattice constants are already close to experiment, does adding machine learning provide enough improvement to justify the extra model?

Unlike Chapter 1, you will not run another VASP calculation. Instead, you will work with a provided dataset of cubic materials containing DFT/PBE lattice constants, experimental lattice constants, simple chemistry descriptors, structure-family labels, and a fixed train/test assignment.

Why This Problem Comes Second

Chapter 1 starts from a property where PBE can show a large material-dependent error. Chapter 2 intentionally starts from a property for which DFT is often already quite good. This changes the scientific question from “Can ML correct a large error?” to “Does ML add meaningful value beyond a strong baseline?”

Workflow



Dataset

Column	Meaning
material	Material identifier.
structure_family	Cubic structure family such as FCC, BCC, diamond, zincblende, rocksalt, or perovskite.
pbe_lattice_A	DFT/PBE lattice constant in Å.
experimental_lattice_A	Experimental reference lattice constant in Å.
mean_atomic_number	Simple chemistry descriptor.
electronegativity_range	Simple chemistry descriptor measuring elemental contrast.
split	Fixed train/test assignment so everyone evaluates the same held-out materials.

CHAPTER 2 - WORKFLOW

Step-by-Step: Lattice Constants

1. Plot PBE directly against experiment

Make the same kind of parity plot you used in Chapter 1: PBE on the x-axis, experiment on the y-axis, and the ideal y = x line.

Checkpoint

Before training anything, answer: How large is the raw DFT error? Does PBE appear systematically high, systematically low, or mixed? Is there enough error for ML to have something meaningful to learn?

2. Establish the raw DFT baseline

Calculate the absolute PBE error for each material and the mean absolute error (MAE). This is the performance your ML model must beat.

$$\mathbf{MAE_PBE = mean(|a_PBE - a_exp|)}$$

3. Define the lattice-constant correction

Use the same delta-learning philosophy as Chapter 1:

$$\mathbf{\Delta a = a_exp - a_PBE}$$

The final corrected prediction will be a_PBE + Δa_ML.

4. Use the fixed train/test split

Do not reshuffle the data. The dataset includes a fixed split so the class evaluates the same held-out materials. One-hot encode the structure-family label and use the provided chemistry descriptors plus the raw PBE value as ML inputs.

5. Train a simple regression model

The template suggests a scaled Ridge regression. The goal is not to perform an extensive model search. The goal is to determine whether a simple, regularized correction model can outperform the raw DFT baseline.

6. Compare raw PBE and ML on held-out materials

For each test material calculate both errors. Then make two parity plots using the same axis limits:

Left panel	**Right panel**
Experimental lattice constant vs raw PBE lattice constant.	Experimental lattice constant vs ML-corrected lattice constant.

7. Identify where ML helps and where it hurts

Add a Boolean column such as improved = ML error < PBE error. The model can improve the average error while making one or more individual materials worse. Both observations matter.

SYNTHESIS

What the Two Problems Show Together

	Chapter 1	Chapter 2
Property	Electronic gap / metallicity	Lattice constant
DFT role	Live VASP calculation for two held-out materials	Provided DFT dataset
Typical baseline behavior	Can be highly material dependent; some gaps are substantially underestimated while metallic classification may already be correct	Often already close to experiment
ML target	ΔE_gap = E_exp - E_PBE	Δa = a_exp - a_PBE
Main question	Can ML learn when a large, small, or zero correction is needed?	Can ML improve enough over a strong DFT baseline to be worthwhile?
Scientific lesson	Do not generalize “DFT is bad” from one material/property	Do not assume “ML is better” simply because a model can be trained

Final Questions to Be Able to Answer

- Why is a comparison to experiment necessary even when the DFT calculation is numerically converged?
- Why can the same exchange-correlation approximation be adequate for one material/property and poor for another?
- Why is predicting the DFT-to-experiment residual often more scientifically interpretable than asking ML to predict the experimental value from scratch?
- Why must ML be compared to the raw DFT baseline on held-out data?
- When would you decide not to use the ML correction at all?

End-of-day perspective

A strong computational materials workflow is not “simulation versus ML versus experiment.” It is knowing what each source of information contributes, where each one fails, and how to combine them without losing physical interpretation.